RAINWATER HARVESTING SYSTEMS GUIDANCE FOR SCHOOLYARD APPLICATIONS

Factsheet | HGIC 1729 | **Published:** Sep 8, 2015 | Print

Rainwater harvesting is the collection and storage of rainwater from roof surfaces for use in both potable and non-potable applications, and for stormwater, erosion and flood control. Rainwater harvesting is an ancient practice and is still widely used throughout the world, becoming more popular in residential yards and schoolyards in the United States.

For the purpose of this guidance document, the focus is on the collection and non-potable use of rainwater in schoolyard landscapes.

Why Harvest Rainwater?

Irrigation: Harvested rainwater can be used to irrigate landscape beds, butterfly gardens, rain gardens, and container plants, as well as to create wildlife features such as birdbaths or butterfly puddling areas.

Stormwater Runoff: Rainwater harvesting manages polluted runoff by decreasing the volume of stormwater that moves across the landscape, transporting pollutants, such as fertilizers, pet waste, sediment, and litter, to nearby waterways.

Flooding & Erosion issues: This practice can also be used to manage flooding and erosion around the foundation of a building.

How Much Water Can Be Collected?

As a general rule of thumb, for every one-inch of rain and every one-square foot of roof surface, the potential exists to capture over half of a gallon of water. To put this into perspective, for a one-inch rain event, a 1000 square foot roof can yield more than 600 gallons of water. Rainwater harvesting provides an excellent tool to teach students about local rainfall patterns, water conservation, impervious surfaces and watersheds, as well as the volume of water that falls on a property when it rains.



Did you know? A 1000 square foot roof area can generate 600 gallons of water during a one-inch rain event.

Use of Harvested Rainwater in the Schoolyard: Bacteria and other pollutants (such as fecal matter from a visiting squirrel or bird, or heavy metals from roofing materials) can accumulate on roof surfaces. Because harvested rainwater is collected as water flows off roof areas, these pollutants can be washed off the roof and end up in the collection tank. Due to these potential health concerns, application of harvested rainwater on edibles can only be safely done by following specific protocols; for additional information visit **HGIC 1728, *Best Practices for Application of Harvested Rainwater on Edibles*.**

For the purposes of schoolyard applications where safety is of the upmost importance, the following safety protocols are recommended for application of harvested rainwater:

- Harvested rainwater should only be used on non-edibles (butterfly gardens, native plant gardens, rain gardens, etc.) To avoid risk of using potentially contaminated water on garden edibles.
- Students should wash their hands thoroughly after direct contact with harvested rainwater.
- All rain barrels and cisterns should be labeled “do not drink” at spigot to avoid accidental ingestion.

Rain Barrels

Rain barrels are commonly used to collect rainwater as they are typically easy to find and relatively inexpensive. Rain barrels, which are sized to hold up to 100 gallons of water, are used for small-scale rainwater harvesting.



Water is directed to the rain barrel using gutters, downspout, rain chain or simple sheet flow. Ensure that the point of water entry is screened and secure to keep mosquitoes out.

Getting Water to the Barrel: Rainwater is directed into the barrel using gutters, downspout, rain chain or simple sheet flow from the roof. The point where water enters the rain barrel, typically the top, must be screened to prevent mosquito breeding and to restrict debris such as leaves, twigs and small animals from entering the barrel. This screened entryway should be cleaned regularly to ensure water flow is not restricted. It is very important to securely fasten the lid as a safety precaution; this will prevent small animals from becoming entrapped in the barrel.

Moving Water from the Rain Barrel: Regardless of whether a school has one rain barrel or multiple barrels, the total volume of water is generally too small for a conventional pump; therefore, the rain barrels should be elevated to allow for gravity feed (i.e., using the force of gravity to push water out of the barrel). For every 1 foot of elevation, approximately 0.4 pounds of pressure (PSI) is created. Typically, rain barrels are elevated 12 to 36 inches above the ground, which allows for enough pressure to push water through a spigot to fill a watering can, hose or for drip irrigation. Note: Elevated barrels rarely have enough pressure to force water through a soaker hose, unless a low pressure soaker hose is used. Drip irrigation or hose application is recommended.

The end use for the water should be located as close as possible to the rain barrel system, as this will encourage the use and maintenance of the rain barrel. For example, if the water is being used to irrigate a butterfly garden, it is best to have the rain barrel(s) adjacent to the butterfly garden, and not 40 feet away.

And remember, rain barrels work best when used.



Figure 1. Area elementary school students painted these colorful rain barrels located at the College of Charleston Grice Marine Laboratory. The four rain barrels are “daisy chained.”
Kim Counts Morganello, ©2015 Clemson Extension

Building Storage Capacity: Rain barrels can be connected or “daisy-chained” to increase storage capacity. For example, linking two 50-gallon rain barrels can create a 100-gallon capacity system. See Figure 1.

Elevating Rain Barrels: As previously mentioned, elevating a rain barrel increases the water pressure coming out of the spigot or hose. A raised barrel also allows for easier access to the spigot, which can be beneficial when students are filling a watering can or bucket.

It is imperative to make sure the barrel is stable and secure regardless of the anchoring method. A 55-gallon rain barrel weighs over 450 pounds when full; therefore, elevated rain barrels need to be secured to prevent injury. Techniques for elevating rain barrels include creating a cinder block or paver base, or building a platform that elevates the barrel off of the ground. Anchoring the rain barrels to the stand using metal strapping can also be used to stabilize the barrel. See Figure 2.



Figure 2. Both of these rain barrels are elevated using a wooden stand to provide head pressure so that water will move from the barrel to the point of use. The barrels are either enclosed or strapped to the wooden stand for stability and safety purposes.
Kim Counts Morganello, ©2015 Clemson Extension

Key Features of a Rain Barrel:

- Spigot that can be turned on and off.
- Emergency overflows that allow water to escape when barrel is full; typically flexible pipe or PVC that directs water overflow away from the foundation, doorways, sidewalks, etc.
- Dark color to prevent sunlight penetration and algal growth.
- Recycled barrels should be food grade and have never been used to transport chemicals.
- Secure point of water entry.
- Screened entry and exit points to limit mosquitoes.

To learn more about rain barrel maintenance and construction, visit **www.clemson.edu/carolinaclear** and download a free copy of the *Rainwater Harvesting for Homeowners Guide* produced by Clemson Extension Carolina Clear.

Cisterns

Rainwater can be harvested on a larger scale through the use of cisterns. A cistern is a storage tank that can hold 100 gallons or more. Cisterns come in all shapes and sizes and can be made of a variety of materials including plastic and metal. Cisterns can be installed either above or below ground.



Figure 3. A 900-gallon cistern located at PACE Academy in Charleston, SC. The first flush diverter, depicted on left hand side of photo, diverts the first flush of water, which contains the most debris and potential pollutants. Kim Counts Morganello, ©2015 Clemson Extension

Getting Water to the Cistern: Cisterns tend to be more complex than rain barrels because additional emphasis is placed on filtering the water. This pre-filtration can be done by including downspout, basket, or vortex filters and/or a first flush diverter. The “first flush,” or the initial rainfall coming off a roof, is the dirtiest water produced during a rain storm. Using a first flush diverter simply diverts this water away from the cistern before it has a chance to enter the tank. See Figure 3.

Maintaining the pre-filtration devices is critical to assuring the integrity of cisterns, as cisterns cannot be cleaned as easily as rain barrels. For best results, use the water regularly and ensure baskets and screens are cleaned (typically once a month) and empty first flush diverter basket filters (typically once a month).

Moving Water from a Cistern: Water in a cistern can be distributed using gravity feed or a pump, and can be set on a timer. When using a pump, pressure will resemble that of municipal water supplies and water can be transported over greater distances than when relying on gravity feed. If the system is set to run automatically, it is important to include an automatic shut-off, typically a float switch; otherwise, the pump may overheat if the tank is empty.

To learn more information, including potential training opportunities, visit the American Rainwater Catchment Systems Association www.arcsa.org or visit the Clemson University Center for Watershed Excellence at **www.clemson.edu/watershed**



Figure 4. A 1,100-gallon cistern located at the Clemson University Coastal Research and Education Center in Charleston, SC. This system is set on a timer to deliver water to a drip irrigation system that waters adjacent landscaped beds and greenhouse.
Kim Counts Morganello, ©2015 Clemson Extension

See Figure 4.

Maintenance: Routine system maintenance is recommended at a minimum of twice a year for both rain barrels and cisterns. Maintenance needs should be communicated to appropriate individuals. The best maintenance is regular use of the rainwater harvesting system allowing for optimal water quality and system function.

Maintenance Check List:

Rain Barrels:

- Clear debris from gutters and downspouts.
- Check rain barrel is secure.
- Check top is secure.
- Check water for algae, insects and other forms of life and drain if needed.
- Check spigot function.
- Clear any debris from screening at point where water enters the tank.
- Check overflow pipe for signs of erosion and redirect away from building if needed.

Cisterns:

- Clear debris from gutters and downspouts.

- Clear debris from pre-filtration devices (downspout filter, first flush diverter, etc).
- Check cistern is secure.
- Check cistern top/access is secure.
- Check water for algae, insects and other forms of life and drain if needed.
- Check spigot function.
- Check overflow for signs of erosion and adjust (ex. add river rock) if needed.

Rain Gardens

Regardless of the size of the collection tank, ideally overflow can be directed to an adjacent landscaped area or rain garden. Rain gardens should be considered part of the rainwater harvesting system. Rain gardens are landscaped depressions that allow water to pool and infiltrate, thus reducing the amount of stormwater runoff. The plants and microbes in the soil do the heavy lifting of removing pollutants from stormwater runoff. To learn more about rain gardens, visit **www.clemson.edu/carolinaclear** and download a free copy of the *Carolina Clear Rain Garden Manual*. See Figure 5 & 6.



Figure 5. Small cistern with overflow directed to an adjacent rain garden-located at SC Department of Natural Resources Fort Johnson Marine Center, James Island, SC.
Kim Counts Morganello, ©2015 Clemson Extension



Figure 6. The overflow of this rain barrel is directed to a nearby rain garden at the Dorchester County Government Building in Summerville, SC.
Kim Counts Morganello, ©2015 Clemson Extension

Harvesting rainwater is a fun and educational way to conserve water and practice good watershed stewardship. Although this water is not treated, thus not drinkable, it may still be safely applied in a schoolyard setting by following these recommended application, design and maintenance guidelines. To learn more, visit **www.clemson.edu/carolinaclear**.

Originally published 09/15

If this document didn't answer your questions, please contact HGIC at **hgic@clemson.edu** or 1-888-656-9988.

Author(s)

Kimberly C. Morganello, Water Resources Associate, Clemson University

This information is supplied with the understanding that no discrimination is intended and no endorsement of brand names or registered trademarks by the Clemson

University Cooperative Extension Service is implied, nor is any discrimination intended by the exclusion of products or manufacturers not named. All recommendations are for South Carolina conditions and may not apply to other areas. Use pesticides only according to the directions on the label. All recommendations for pesticide use are for South Carolina only and were legal at the time of publication, but the status of registration and use patterns are subject to change by action of state and federal regulatory agencies. Follow all directions, precautions and restrictions that are listed.

Was this helpful?

YES NO

PREVIOUSBest Practices for Application of Harvested Rainwater on Edibles